ATENEO DE MANILA
UNIVERSITY
Loyola Schools

SYLLABUS FOR UNDERGRADUATE COURSES
MAJOR, CORE CURRICULUM and ELECTIVES
Student Copy

A. COURSE INFORMATION

COURSE NUMBER	ENGG 151.02	**NO. OF UNITS**	1 unit
COURSE TITLE	Signals, Spectra, and Signal Processing, Laboratory		
PREREQUISITE/S	Prerequisites: ENGG 27.01, ENGG 27.02 Co-requisite: ENGG 151.01		
DEPARTMENT/ PROGRAM	Department of Electronics, Computer, and Communications Engineering	**SCHOOL**	School of Science and Engineering
SCHOOL YEAR	2025-2026	**SEMESTER**	Second
INSTRUCTOR/S	Luisito L. Agustin		
VENUE/PLATFORM Canvas https://canvas.ateneo.edu/	**SECTION** ENGG 151.01 A ENGG 151.02 LAB2-KL	**SCHEDULE**	M-TH 0800-0930 CTC 403 F 1000-1200 CTC 403

In addition to official class hours, students are expected to log at least one more hour per week in the lab, during "open lab" hours, towards meeting the learning objectives of the course.

DSP lecture and laboratory courses never need to synchronize with each other.

This syllabus contains one appendix:
Appendix 1. Attestation

To participate in the course, a student is required to sign the attestation in the appendix at the end of this syllabus and submit a copy of the entire syllabus with the signed attestation to the instructor.

B. COURSE DESCRIPTION

WHERE IS THE COURSE SITUATED WITHIN THE FORMATION STAGES IN THE FRAMEWORK OF THE LOYOLA SCHOOLS CURRICULA	
	FOUNDATIONS: Exploring and Equipping the Self
	ROOTEDNESS: Investigating and Knowing the World
*	DEEPENING: Defining the Self in the World
	LEADERSHIP: Engaging and Transforming the World

ENGG 151.01
SIGNALS, SPECTRA, AND SIGNAL PROCESSING, LECTURE
3 units
Prerequisites: ENGG 27.01, ENGG 27.02
Co-requisite: ENGG 151.02

This is a lecture course on signals, spectra, and signal processing. The course starts with a discussion of the nature of digital signals resulting from sampling and quantization processes. Linear time-invariant (LTI) systems are introduced as the most general linear systems for processing one-dimensional single channel signals in the time domain while Fourier transforms allow analysis of signals in the frequency domain. Important properties of LTI systems are studied both in the time domain and the frequency domain through z-transforms. Applications involving the processing of real-world digital signals such as digital audio, images and video are also studied.
(Undergraduate Bulletin of Information, 2020)

ENGG 151.02
SIGNALS, SPECTRA, AND SIGNAL PROCESSING, LABORATORY
1 unit
Prerequisites: ENGG 27.01, ENGG 27.02
Co-requisite: ENGG 151.01

This laboratory course complements the topics discussed in Signals, Spectra, and Signal Processing, Lecture. The course starts with a discussion of the nature of digital signals resulting from sampling and quantization processes. Linear time-invariant (LTI) systems are introduced as the most general linear systems for processing one-dimensional single channel signals in the time domain while Fourier transforms allow analysis of signals in the frequency domain. Important properties of LTI systems are studied both in the time domain and the frequency domain through z-transforms. Applications involving the processing of real-world digital signals such as digital audio, images and video are also studied.
(Undergraduate Bulletin of Information, 2020)

ENGG 151.01 and ENGG 151.02 address the need for courses on *Digital Signal Processing* for BS Computer Engineering and *Signals, Spectra, Signal Processing* for BS Electronics Engineering specified by CHED (Commission on Higher Education) in CMO 87 and CMO 101 (CHED Memorandum Order 87 for BS Computer Engineering and CHED Memorandum Order 101 for BS Electronics Engineering).

C. COURSE LEARNING OUTCOMES and PROGRAM LEARNING OUTCOMES

Note: Details are subject to change.

The Program Educational Objectives and the Program Learning Outcomes of BS Computer Engineering and BS Electronics Engineering are available through this link (https://docs.google.com/document/d/1BCH-i_qqPR4DikXxxO3kIhgy982I0qLU/edit) .

Program Learning Outcomes (PLO)

Upon graduation, our graduates are expected to:

a) Use mathematical, scientific and technical concepts to address complex engineering challenges.
b) Design, organize and conduct experiments, and evaluate data.
c) Design a system, component, or procedure that satisfies requirements and meets standards, taking into account practical limitations such as those pertaining to the economy, the environment, ethics, health and safety, sustainability and/or other socio-cultural constraints.
d) Function in multidisciplinary groups.
e) Recognize, define and address complex engineering challenges.
f) Exercise ethical and professional responsibility.
g) Communicate effectively.
h) Determine how engineering solutions affect society, the environment, the economy, and the world at large.
i) Acknowledge the importance of and possess the capacity for lifelong learning.
j) Understand current and contemporary affairs.
k) Apply the methods, skills, and modern instruments necessary for engineering practice.
l) Practice proficient engineering and management skills to lead and collaborate in teams and projects in multidisciplinary settings.
m) Understand a specialized field of professional practice.

By the end of this course (ENGG 151.01 and ENGG 151.02), students should be able to:

COURSE LEARNING OUTCOMES	PROGRAM LEARNING OUTCOMES (see link)												
	a	b	c	d	e	f	g	h	i	j	k	l	m
ENGG 151.01 and ENGG 151.02	D	E			E								
CLO1: discuss signal processing		E			E								
CLO2: do computations with signals	D												
CLO3: carry out computations relating to LTI systems	D												
CLO4: carry out computations relating to frequency analysis	D												

Legend: D – Demonstrative; E – Enabling; I - Introductory

The CHED course description for an equivalent course states that upon completion of the course, the student must be able to conceptualize, analyze and design signals, spectra and signal processing systems.

The course will pursue learning outcomes that are feasible and reasonable, based on current interpretations of the course description, and subject to all feedback that may be received as the course progresses.

D. COURSE OUTLINE and LEARNING HOURS

Note: Details are subject to change.

Course Outline	CLOs	Estimated Contact or Learning Hours
introduction, basic concepts, signals and systems	1, 2, 3, 4	3 to 6
correlation	1, 2	15 to 18
linear time-invariant (LTI) systems z-transforms frequency domain analysis of LTI systems	1, 2, 3, 4	15 to 18
frequency analysis of signals discrete Fourier transform Fast Fourier Transforms	1, 2, 3, 4	15 to 18

Tentative list of lab exercises
* Lab Exercise 1 - Peak Sample Values
* Lab Exercise 2 - Quantization
* Lab Exercise 3 - Signal Generator using Digital Sinusoidal Oscillators
* Lab Exercise 4 - Fourier Analysis: Musical Notes from Video Files
* Lab Exercise 5 - Magnitude Response of a Biquad Filter
* Lab Exercise 6 - Fast Fourier Transform on a Spreadsheet

The course will cover topics that can feasibly and reasonably be covered, based on current interpretations of the course description, and subject to all feedback that may be received as the course progresses.

calendar

Note: Details are subject to change.

missing item: Faculty Day, historically held on a Friday

	Lecture		Lab
	Mon	Thu	Fri
week 0	Jan 5 Jan 7 - 2025-2 start	Jan 8	Jan 9
week 1	Jan 12	Jan 15	Jan 16
week 2	Jan 19	Jan 22 Project 1 start	Jan 23 Lab 1 start
week 3	Jan 26	Jan 29 progress report due	Jan 30 progress report due
week 4	Feb 2 President's Day	Feb 5 progress report due	Feb 6 Lab 1 due Lab 2 start
week 5	Feb 9	Feb 12 progress report due	Feb 13 progress report due
week 6	Feb 16 early work due	Feb 19 Project 1 due Project 2 start	Feb 20 Lab 2 due Lab 3 start
week 7	Feb 23	Feb 26 progress report due	Feb 27 progress report due
week 8	Mar 2	Mar 5 progress report due	Mar 6 Lab 3 due Lab 4 start
week 9	Mar 9	Mar 12 progress report due	Mar 13 progress report due Lab 5 start
week 10	Mar 16 early work due	Mar 19 Project 2 due Project 3 start	Mar 20 Eid'l Fitr ?
week 11	Mar 23	Mar 26 progress report due	Mar 27 Lab 4 due progress report due
week 12	Mar 30 Holy Week Break	Apr 2 Holy Week Break	Apr 3 Holy Week Break
week 13	Apr 6	Apr 9 Valor Day	Apr 10 Lab 5 due Lab 6 start
week 14	Apr 13	Apr 16 progress report due	Apr 17 progress report due
week 15	Apr 20 Thesis Week Break	Apr 23 Thesis Week Break Thesis Presentations Apr 23 - 24 progress report due	Apr 24 Thesis Week Break Thesis Presentations Apr 23 - 24 Lab 6 due
week 16	Apr 27 early work due	Apr 30 Project 3 due	May 1 Labor Day
week 17	May 4	May 7	May 8 study day
week 18	May 11 finals	May 14 finals	May 15 *May 16 - 2025-2 end*

Homework may be assigned each class day.

E. ASSESSMENTS AND RUBRICS

Assessment Tasks	Assessment Weight	CLOs
lab exercises (each graded over 100 points)	90%	1, 2, 3. 4
homework (homework and all other tasks that are not components of the lab exercises, all earning "homework points" whose cumulative value will be the basis for the "homework grade")	10%	1, 2, 3. 4

Each lab exercise will typically have several components that will be graded separately:

Preliminary Work:
Preliminary Implementation (30%)

Final Submission (due on the same date):
Final Implementation (30%)
Lab Report (40%)

The lab grade for an exercise is obtained by adding the points earned from the three components. No single component will have a weight of more than 20% of the total course work. This means no single submission qualifies as a "major requirement".

RUBRICS:

The following are general rubrics for projects and similar asynchronous activities. Where feasible, more detailed rubrics may be given for specific tasks.

F	
D	General specifications are met.
C	
B	All general and detailed specifications are met.
A	excellence beyond meeting all general and detailed specifications

Indicators of excellence include, but are not limited to, meticulous attention to detail, significant insights, efficiency, neatness, regularity of submission, early submission, careful documentation, coherence, organization, clarity, conciseness, ability to integrate material from outside the course and originality of ideas.

F. TEACHING and LEARNING METHODS

TEACHING & LEARNING METHODS and ACTIVITIES	CLOs
readings, lecture notes	1, 2, 3, 4
programming assignments and related homework and reports	1, 2, 3, 4
major software projects and related reports	1, 2, 3, 4
video conferencing	1, 2, 3, 4

G. REQUIRED RESOURCES

[Proakis]
(book) John Proakis and Dimitris Manolakis: Digital Signal Processing, 4th ed, 2007.

[Walpole]
(book) Walpole, et al: Probability and Statistics for Engineers and Scientists, Pearson, 9th ed, 2012.
(we need the material on correlation)

[CodeBlocks]
(software) The Code::Blocks Team: *Code::Blocks Integrated Development Environment (IDE) for C++*, latest version- Code::Blocks 25.03, March 31, 2025, https://www.codeblocks.org/.
(or an appropriate alternative)

[Calc]
(software) The Document Foundation: LibreOffice Calc Spreadsheet, ver. 25.2.3.2, 2025, www.libreoffice.org.
(or an appropriate alternative)

[VLC]
(software) VLC Media Player, ver. 3.0.21, 08-Jun-2024, https://www.videolan.org/vlc/.
(or an appropriate alternative)

[Audacity]
(software) Audacity audio editor, ver. 3.7.4, 0 Jun 2025, https://www.audacityteam.org/download/.
(or an appropriate alternative)

H. SUGGESTED RESOURCES

[Audacity]
(software) Audacity audio editor, ver. 3.7.4, 0 Jun 2025, https://www.audacityteam.org/download/.

[CodeBlocks]
(software) The Code::Blocks Team: *Code::Blocks Integrated Development Environment (IDE) for C++*, latest version- Code::Blocks 25.03, March 31, 2025, https://www.codeblocks.org/.

[Calc]
(software) The Document Foundation: LibreOffice Calc Spreadsheet, ver. 25.2.3.2, 2025, www.libreoffice.org.

[VLC]
(software) VLC Media Player, ver. 3.0.21, 08-Jun-2024, https://www.videolan.org/vlc/.

(online video) Veritasium: *The Most Important Algorithm Of All Time*, https://www.youtube.com/watch?v=nmgFG7PUHfo, Nov 3, 2022, Veritasium (https://www.youtube.com/@veritasium).

[Ida] (book) Nathan Ida: Sensors, Actuators, and Their Interfaces: A Multidisciplinary Introduction, 2nd ed, IET, London, 2020.

[Walpole]
(book) Walpole, et al: Probability and Statistics for Engineers and Scientists, Pearson, 9th ed, 2012.
(we need the material on correlation)

(book) Alan V. Oppenheim and Ronald W. Schafer: *Discrete-Time Signal Processing*, 3rd ed, Pearson, 2009.

[Proakis]
(book) John Proakis and Dimitris Manolakis: Digital Signal Processing, 4th ed, 2007.

(book) Sanjit K. Mitra : *Signal Processing – Computer Based Approach*, 2nd ed, McGraw-Hill, 2005.

(book) Steven W. Smith: *The Scientist and Engineer's Guide to Digital Signal Processing*, 1997.
(may be downloaded at http://www.dspguide.com/)

(book) Craig Marven and Gillian Ewers: *A Simple Approach to Digital Signal Processing*, Texas Instruments, 1996.

(book) John Proakis and Dimitris Manolakis: *Digital Signal Processing*, 3rd ed, 1996.

(journal article) James W. Cooley and John W. Tukey: *An algorithm for the machine calculation of complex Fourier series. Math. Comput. 19 (90): 297–301, 1965 doi:10.2307/2003354. JSTOR 2003354.*

I. GRADING SYSTEM

Each lab exercise earns a grade in the range [0..100].
L: average grade of lab exercises

Homework points are earned from homework.
H: homework grade in the range [0..100] based on total homework points earned

CS: class standing in the range [0..100] = 0.9 *L*+ 0.1 *H*
The final grade is obtained from *CS* as follows:
[0,60) F, [60,68) D, [68,74) C, [74,80) C+, [80,87) B, [87,93) B+, [93,100] A.

J. CLASS POLICIES

J.1. Policies of the Ateneo College

All relevant policies of the Ateneo College apply. These policies include, but are not limited to, the following:

Higher Education Gender Policy

Code of Decorum and Administrative Rules on Sexual Harassment, Other Forms of Sexual Misconduct, and Inappropriate Behavior

Undergraduate Student Handbook, 2025 edition

(links as provided in a memo dated 09 December 2025 by the Office of the Assistant Vice President for Undergraduate Education)

The Code of Academic Integrity is an integral part of the student handbook and requires that: *Students should ensure that all submitted work, both individual and group, is the product of their own actions, reflection and learning.* The rule applies without exception to all submitted work including those where generative AI may have been used.

J.2. Files, Documents, Communications and Submissions

Fonts

12 pt. Liberation Serif - base font for all documents
12 pt. LIberation Sans - alternative base font
`10 pt. Liberation Mono or 12 pt. Liberation Mono`
`- to be used where an equispaced font is desired, such as for C++ code`

Documents

Word processor documents
pages: A4 with 2 cm. margins on all sides
Export all word processor documents to pdf.

C++ files
Use .cpp extensions for implementation files, .h extensions for header files. Do not use any other extension.

All files and documents submitted must contain the names of the authors.

Submissions

Submit thru the submission link for a requirement in Canvas whenever possible. Submission of a requirement thru Canvas may be done only once. Submit thru official Ateneo email if submission thru Canvas is not possible or if a requirement needs to be resubmitted. Do not submit a requirement as an attachment to a message or comment in Canvas. Requirements submitted as attachments to messages or comments in Canvas are not valid submissions and will not be graded. A requirement submitted via email overrides any submission of the same requirement thru Canvas regardless of submission time. Where multiple versions of a requirement are submitted via email, only the latest will be graded, regardless of the quality of earlier submissions.

pdf - to be used for all formatted documents
txt - to be used for plain text documents
zip - to be used when more than one file is submitted or when submitting other types of files

Requirements must be submitted in full at all times, including cases where the intended submission is partly or fully the same as a previous submission. In particular, major requirements must be submitted completely, regardless of whether the submission, or parts of it, may be the same as in a previous early work submission or progress report.

Links will not be accepted as submissions. Actual documents should be submitted. Where applicable, submissions must conform to specified file types, naming conventions and related specifications.

Students are responsible for ensuring that requirements they submit are what they intended to submit. Submission of requirements that turn out to be defective or inaccessible is equivalent to non-submission of requirements.

For submissions involving several files, collect the files to be submitted, one at a time into a separate folder. Make sure the files are complete and that all files are relevant to the submission. Compress the files into a single zip file. For C++ projects, avoid including executables, DLLs and object files (*.exe, *.dll, *.o, *.obj). For project files from an online repository, do not include extra files from the repository, such as those that might be needed to track versions of files.

For software projects, the inclusion of executable files is prohibited unless explicitly requested. The presence of two versions of a file in a single submission causes confusion and should be avoided. The penalty for including executables, DLLs, extraneous repository files and/or irrelevant files in a submission is a scaling of whatever score is earned from the requirement by 80%.

Communication

Email using official email addresses, *not* coursed thru the LMS, is preferred (Support for email threads is better than if coursed thru the current LMS). Include the course number, your surname, and the course item of concern in the subject line of your email,
e.g.

ENGG 456 Agustin hw 987 clarification

Penalties

A scaling factor of 0.8 will be applied to points earned from submissions containing invalid files. e.g.

executable files
files unrelated to the course requirement
docx or odt files included in the submission (Export all word processor documents to pdf).
a docx file whose extension has been changed to pdf
invalid zip file, such as a rar file whose extension has been changed to zip
documents using fonts unjustifiably deviating significantly from those listed
multiple versions of a document
document not containing name of author

The score is 0 for:

links submitted instead of actual files (equivalent to non-submission)
submissions that are defective or inaccessible (equivalent to non-submission)

J.3. Deadlines and Related Matters

Deadlines for all course requirements will be during official class time. In cases where a deadline has a grace period or provision for late submission the modified deadline does not have to fall during official class time but must be pegged to the original deadline that was scheduled during official class time. These deadlines pegged to the original deadline during official class time may fall on holidays or no-class days.

Requirements assigned at least three days before they are due must be submitted on the day before they are due, to provide an allowance for seeking remedies in case problems arise in the initial attempt to submit a requirement. If problems arise in the initial attempt, students must document proof that problems arose, and must seek remedies so that the requirement is submitted on time. Students must verify that requirements have been successfully submitted and be able to show proof that the submission completed successfully in case of disputes. While submissions will be valid up to the actual deadline, students who complete requirements very close to the deadline do so at their own risk. Poor connectivity and malfunctioning devices are not valid reasons for missing deadlines.

The grade is 0 for requirements not submitted on time. No make-up work will be given for a requirement not submitted on time. Students are responsible for scheduling submission times so that deadlines will not be missed, including scheduling submissions much earlier than the deadline if internet connectivity is a concern. A submission must be completed -- *not started* -- before the deadline. A student who misses a submission deadline, then emails the requirement because the deadline was missed, will not get credit for the submission thru email.

A 24-hr late submission window will be implemented for major requirements. Points earned from major requirements submitted late for whatever reason will be scaled by 90%. Grades will be zero for requirements not submitted before the end of the late submission window. Grades will be zero for requirements whose submission was started before the end of the late submission window but did not complete before the end of the late submission window.

All requirements are submitted individually. Where group work is allowed the submission of one student may be on time even if a groupmate submits late or fails to meet the deadline (and vice versa).

J.4. Course Work

Summary of Grading Procedure for Course Work

1. The course work or requirement is evaluated based on its merits, possibly using available rubrics, point systems or equivalent objective or subjective criteria. Where late submission is not allowed, course work submitted late need not be evaluated.
2. Relevant quality factor penalties are applied.
3. Relevant time factor penalties are applied.
4. Relevant penalties for large groups and unauthorized collaboration are applied.
5. If there are suspected offenses involving dishonesty, section J.1 applies.

Authorship

Students must claim authorship of all components of all course work they submit by placing their names and those of their groupmates (where group work is allowed) on each course work component.

In general, names should be placed on the upper left. Hence, names are expected on the upper left of a pdf document or a circuit diagram, the first few lines of source files or the uppermost leftmost cell of a spreadsheet.

For programming projects, names are not required on files generated by code worked on.

Quality Factor Penalties

Quality is in doubt if authors do not bother claiming authorship. Course work that include components whose authorship is not documented will be penalized 20% of points earned. For group work, the course work of each group member will be penalized 20% of points earned if there are irregularities in reporting group composition by any one group member.

Other more specific quality factor penalties may apply to different types of course work.

Projects and Group Work

All requirements are submitted individually. No submission by any one person will be considered as a valid submission for another.

Students are *not* required to work in groups. Where groups are allowed, groups may be composed of at most 3 students and collaboration between and among groups, whether in the past, present or future, is not allowed. The composition of groups may change from one requirement to the next. Requirements where group work is allowed may contain components where collaboration is not allowed. It is reasonable for students working in groups to submit work containing components that are similar or exactly the same as those submitted by groupmates. Students working in groups may submit work containing components different from the result of group collaboration.

For requirements where group work is allowed, all group members are responsible for reporting the composition of a group accurately. Where progress reports are required for a requirement where group work is allowed, the group composition must be reported accurately on the first such progress report and in all subsequent submissions. No groups for the requirement may be formed after the first such progress report. Any irregularity or inconsistency in the reporting of group composition by any member of the group will result in a 20% quality factor penalty on the score earned by each group member for the requirement. The course work of groups formed after the first progress report will be subject to penalties for large groups and unauthorized collaboration.

If there is no issue involving a large group size, unauthorized collaboration or offenses involving dishonesty, project scores are first given based on the specifications for the project according to available rubrics then subjected to a time factor and a quality factor to obtain the project grade:

project grade = time factor X quality factor X project score

Time Factor

	Time Factor
Project is submitted on time or within the grace period.	100%
Project is late. (The time factor will be 80% if early work submission is not a requirement; 90% if early work credits are part of the point distribution.)	90%
Project is not submitted within the allowed time.	0

The penalty for late submission is chosen to be small enough to still allow a good above-average B grade but large enough to discourage the use of late submission as a strategy for gaining an unfair advantage over those who submit on time.

Quality Factor Penalties

The quality factor is 100% unless penalties are incurred from violations. Penalties up to a maximum of 20% are subtracted from 100% . The quality factor is always within the range [80%, 100%]. The penalties are imposed for failure to apply good practices. Capping the penalties at 20% implies that projects with excessive quality violations cannot be given an excellent A but may still qualify for a good above-average B grade depending on whether they meet the actual project specifications.

See also ASSESSMENTS AND RUBRICS**.**

The project grade will be 0 if more than 10% of the project may be found on the internet or the worldwide web.

Laboratory Exercises

Laboratory exercises will be worked on individually and submitted individually. Laboratory exercises will be worked on asynchronously.

Preliminary work on a lab exercise will be due at the end of a class period subject to a 24-hour grace period to accommodate students who may stjill be working on the exercise at the end of the class period. Although the grace period is 24 hours, it is expected that the preliminary work be submitted soon after the class period and nowhere near the end of the grace period. While submissions will be valid up to the actual end of the grace period, students who complete requirements very close to the deadline do so at their own risk. Poor connectivity and malfunctioning devices are not valid reasons for missing the deadline.

The grade is 0 if the submission is not completed by the end of the grace period.

Penalty for Large Groups and Unauthorized Collaboration

In cases where students form project groups of more than 3, or collaborate outside their group (as manifested by unjustified similarities in project implementation), or collaborate on tasks that are supposed to be done individually, the score of each student will be scaled by

$$\frac{min(\text{allowed group size}, max(\text{all group sizes}))}{N}$$

where N is the total number of students in the past and present having similar submissions,

Various utilities may be used to gauge similarity of work. Examples include Turnitin, WinMerge, and Moss.

sample computations:

1. Two groups of two students and three individuals in the present semester submit implementations containing similar components to a past submission by a group of two students.

$\text{allowed group size}=3$
$max(\text{all group sizes})=2$
$min(\text{allowed group size}, max(\text{all group sizes}))=min(3,2)=2$
$N=2\times2+3+2=9$

The score of each student will be scaled by $\frac{2}{9}$.

2. Students form a group of 4 and work on their own implementation.

$\text{allowed group size}=3$
$max(\text{all group sizes})=4$
$min(\text{allowed group size}, max(\text{all valid group sizes}))=min(3,4)=3$
$N=4$

The score of each student will be scaled by $\frac{3}{4}$.

3. Three groupmates submit progress reports with the same content (Progress reports are supposed to be prepared individually).

$\text{allowed group size}=1$
$max(\text{all group sizes})=1$
$min(\text{allowed group size}, max(\text{all group sizes}))=min(1,1)=1$
$N=3$

The score of each student will be scaled by $\frac{1}{3}$.

4. Four students collaborate on a lab exercise component (Lab exercises are done individually).

$$\text{allowed group size} = 1$$
$$max(\text{all group sizes}) = 1$$
$$min(\text{allowed group size}, max(\text{all group sizes})) = min(1, 1) = 1$$
$$N = 4$$

The score of each student will be scaled by $\frac{1}{4}$.

C++ Projects

Course work done using C++ should follow guidelines provided.

In addition, C++ source files and header files must each start with comments documenting the filename and the authors and group members submitting the project.

For C++ programming projects, the penalties in the following table apply. The list includes penalties imposed for failure to apply good programming practices that may not necessarily be directly related to project specifications but improve the quality of code in general.

violation	quality factor penalty
use of global variables	20%
use of `goto`	20%
use of `float` instead of `double`	20%
uninitialized variable	20%
inclusion of executables in submission	20%
failure to document authorship of source files and/or header files; irregularity in reporting group authorship	20%
misuse of file inclusion or header files, or violation of file naming conventions	10% per file containing a violation
failure to use `new` for dynamic memory allocation	2% per violation
non-English and/or inappropriate comments and/or identifier names and/or file names	1% per violation
improper indentation or alignment	1% per violation

The maximum quality factor penalty is 20% regardless of the number of violations.

penalies for misuse of file inclusion or header files, or violation of file naming conventions

Penalize a source file if its extension is not .cpp or .h
Penalize a header file or implementation file with the wrong extension
Penalize a source file that's #includes an implementation file
Penalize a header file that contains an implementation of any routine..

penalties for improper indentation or alignment:

1% penalty for each tab character used to indent; 1% penalty for each pair of curly braces that are not aligned vertically or do not align vertically with the first character of the appropriate keyword; 1% penalty for each line of code that is indented differently from the majority or plurality of lines in a block...

Note: The scaling factor of 0.8 in J.2. is a quality factor penalty.

J.5. Attendance

Attendance is checked towards the start of class. Absence when attendance is checked is considered a cut. Lateness is considered a cut. Students are responsible for monitoring their cuts on their own. Dean's listers are not allowed unlimited cuts.

J.6. Attire

Slippers are inappropriate attire. Students who come to class in slippers or similar footware that are unstrapped at the back of the heal will be penalized increasing multiples of 5 minor points (homework points in the case of lab courses) for each violation, starting with a penalty of 5 minor points for the first violation. If penalties result in minor points less than zero at the end of the semester, the number of points will be set to zero.

J.7. Data Privacy

Personal data may be requested from the students for facilitating the delivery of the course. Personal data collected from students, and that are not available from anywhere else, will be kept confidential and will not be disclosed to other parties without consent of the student.

J.8. Supplementary Policies

Supplementary policies for the course may be issued if needed. These will become part of the syllabus.

J.9. Language

This is an English language course. All components of all course requirements will be in English.

J.10. Participation in the Course

To participate in the course, a student is required to sign the attestation in the appendix at the end of this syllabus and submit a copy of the entire syllabus with the signed attestation to the instructor.

J.11. Exceptions

Exceptional cases may arise. The burden of proof is generally on whoever is claiming the exception.

K. CONSULTATION HOURS

NAME OF FACULTY	EMAIL	DAY/S	TIME
Luisito L. Agustin	lagustin@ateneo.edu	W-Th	1000-1200 or by appointment

L. ADDITIONAL NOTES

This syllabus contains one appendix:

Appendix 1. Attestation

Appendix 1. Attestation

To participate in the course, you are required to sign the attestation below and submit a copy of the entire syllabus with your attestation to your instructor:

I hereby attest that I have read this_ENGG 151.02 **_syllabus and agree to be responsible for abiding by all its provisions.**

Peter Julius M. Estacio	Jan. 18, 2026
Signature over complete name	**Date Signed**